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LIGHT AND HUMAN HEALTH: AN OVERVIEW OF THE IMPACT OF OPTICAL RADIATION ON VISUAL, CIRCADIAN, NEUROENDOCRINE, AND NEUROBEHAVIORAL RESPONSES

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1.0 Introduction

Light is currently defined as optical radiation entering the eye that provides visual sensation in humans.[1] Despite this specific vision-related definition, light has been increasingly related to a range of ocular circadian, neuroendocrine, neurobehavioral, and therapeutic responses in humans. Technically, the term *optical radiation* should be used to describe the portion of the electromagnetic spectrum spanning ultraviolet, visible, and infrared radiation that stimulates all these biological responses. Since the term light (or lighting) is so widely used to describe optical radiation in the lighting community as well as in the biological and medical research community, the reader is advised that these terms are often used interchangeably in these communities. In this document the term *optical radiation* is used when referring to biological responses other than the visual ones.

Most publications from the Illuminating Engineering Society (IES) to date have described light's impact on the human visual system. In some instances, such as in *The Lighting Handbook*, 10th ed., certain physiological and psychological effects of optical radiation are also discussed. The neurophysiology and neuroanatomy of the human visual system are well documented, and almost all lighting technologies, standards, measurement devices, and applications have (until now) been based solely on that understanding. In this Technical Memorandum, however, the Light and Human Health Committee summarizes the extant body of knowledge on a topic that is new to the architectural lighting community: the impact of optical radiation on circadian, neuroendocrine, and neurobehavioral systems. This document is not concerned with the impact of optical radiation on the skin or other tissues, only ocular exposures.

Much like the dual functions (audition and balance) long associated with the ear, the mammalian eye has dual roles in detecting optical radiation for image-formation (vision) as well as for circadian, neuroendocrine, and neurobehavioral responses. Although this is not a new area of scientific investigation, it is a relatively new topic for many IES members.

Most published scientific research in this area has come primarily from controlled laboratory and clinical experiments, although some studies have attempted to integrate the results into practical situations where they would be beneficial. Therefore, readers will find that the information discussed in this Technical Memorandum is mostly research-based. Recommended practice and application advice are not included.

Since the effects of optical radiation can be profound for human health and well-being, it is increasingly important for the lighting community to understand the direct biological influences of light/dark cycles. It may be possible to develop new lighting technologies and applications that will have health benefits. It may also be possible to employ existing technologies in novel ways to positively influence human health. The counterpoint is also true: it may be possible to employ existing and new technologies in ways that are deleterious to human health.

In brief, this Technical Memorandum describes the retinal mechanisms involved when optical radiation signals are converted into neural signals (a phenomena called *phototransduction*) for vision and for other body functions. Optical radiation reaching the retina not only impacts on how humans see the world, it also regulates physiology and behavior, both directly and indirectly. This includes acute effects such as suppressing pineal melatonin production, elevating morning cortisol production, increasing subjective alertness, enhancing psychomotor performance, changing brain activation patterns to a more alert state, elevating heart rate, increasing core body temperature, activating pupil constriction, and even stimulating circadian clock gene expression.

Perhaps the most important and long-term effect of optical radiation is its ability to reset the internal circadian body clock and synchronize it to local time. Circadian rhythms are daily rhythms that repeat approximately every 24 hours and are driven by an endogenous clock. Nearly all behavioral and physiological parameters exhibit circadian rhythms, and thus circadian clock synchronization is paramount to the body's efficient and appropriate functioning. The neurobehavioral (e.g., sleep/wake cycle) and neuroendocrine (e.g., hormone production) axes are thus influenced by optical radiation both directly (acute effects) and indirectly, via circadian clocks that drive and coordinate the rhythmicity in these systems.

One of the most exciting discoveries in science during the past decade was the identification of a novel type of photoreceptor in the retina, called *intrinsically photosensitive retinal ganglion cells* (ipRGCs). This discovery led to a series of studies to better understand the characteristics, function, and role of the ipRGCs. Questions that have been studied thus far include how the ipRGCs respond to optical radiation, how they regenerate after an optical radiation stimulus, how they provide input to the higher brain centers driving optical radiation-dependent responses, and how they interact with the classical photoreceptors (rods and cones) to provide such information. The findings of these studies are summarized in this Technical Memorandum, but it is important to note that new studies are being conducted and new knowledge is being added to the literature every month, if not every week. The reader is encouraged to continuously search for the most current information on this topic.

The reader will also find here a section discussing the circadian, neuroendocrine, and neurobehavioral systems and how they respond to optical radiation. The majority of the studies in the literature, and hence the majority of the discussion in this document, investigate the interaction and influence of the circadian system on the neurobehavioral and neuroendocrine responses. It is important, however, to keep in mind that the impact of optical radiation on the neurobehavioral and neuroendocrine responses is not exclusively via the circadian system.

In addition to summarizing the current knowledge in retinal photoreceptors and physiology associated with vision and other neural responses, this Technical Memorandum also discusses the impact of various lighting characteristics (quantity, spectrum, timing, duration, light history) on the visual, circadian, neuroendocrine, and neurobehavioral systems. Current knowledge suggests that the optimal light for vision differs from optical radiation for circadian, neuroendocrine, and neurobehavioral responses.

The Light and Human Health Committee decided that, as a first step to introduce this new topic to IES members and the lighting community, it was important to write a short and focused document summarizing the latest research in the phototransduction physiology of the circadian, neuroendocrine, and neurobehavioral systems, coupled with a discussion of the optical radiation characteristics that can be controlled by a lighting professional, which affect the visual and non-visual systems. Thus, this Technical Memorandum has a twofold mission:

- To explain the physiology by which the retina converts optical radiation signals into neural signals for vision and for circadian, neuroendocrine, and neurobehavioral responses.
- To discuss the responses of these systems to optical radiation stimuli.

The next challenge is to incorporate this new information into standard practice and applications, which will be discussed in the future reports from the Committee. It is felt that the information provided here will serve as the foundation for future IES publications that will be more application oriented.

2.0 Overview of the Retina

The eye can be subdivided into two parts: *ocular* (lens and sclera) and *neural* (retina). The function of the ocular parts is to place a focused, clear image of the outside world on the retina with minimal absorption and scattering, and to filter out damaging optical radiation. The retina, which is an extension of the brain, absorbs optical radiation signals and converts them into electrical signals. The processing of visual, circadian, neuroendocrine, and neurobehavioral responses begins in the retina. Subsequently, photic information is sent to various parts of the brain for further processing.

Ganglion cell axons pass electrical signals to the brain (in brief spikes) via two main output pathways from the eye. For visual responses, electrical signals are transmitted from the retina along the primary optic tract to the visual cortex via the lateral geniculate nucleus (LGN) and along the secondary optic tract to the superior colliculus.[2] For non-visual responses, signals are transmitted to the suprachiasmatic nuclei (SCN), ventral preoptic areas (VPO), subparaventricular zones (sPVZ), intergeniculate leaflet (IGL), and the pretectal areas (PT) via the retinohypothalamic tract (RHT).[3]

2.1 Classical Visual Photoreceptors

The retina contains the photoreceptors that detect optical radiation for vision, circadian, neuroendocrine, and neurobehavioral responses. Photoreceptors contain *photopigments* (light-absorbing proteins) that are maximally sensitive to different electromagnetic radiation wavelengths and that have discrete absorbance spectra, which describe the probability of a photon being absorbed as a function of radiation wavelength.

The shapes of the absorbance spectra are very similar for all opsin-based[i] photopigments. These absorbance spectra can be graphically approximated with a Dartnall nomogram, a single-sensitivity function resembling a parabola but having different peak wavelength sensitivities, depending upon the opsin moiety (*moiety* is a part or portion of a molecule). The wavelength of maximal sensitivity (absorption maximum, λ_{max}) is a consequence of the amino acid sequence of the opsin protein and its interaction with the chromophore (a part of the molecule responsible for absorbing photons).

All vertebrate retinas are composed of three layers of nerve cell bodies (the cell body is the largest part of the cell, where the nucleus and other unique structures responsible for making energy and removing waste are located — see **Figure 1**):

- The outer nuclear layer, where cell bodies of rods and cones are located
- The inner nuclear layer, containing cell bodies of bipolar, horizontal, and amacrine cells
- The ganglion cell layer, where cell bodies of ganglion cells (and some displaced amacrine cells) connect to ganglion cells

[i] Opsins are light-sensitive protein-coupled receptors found in photoreceptor cells of the retina.

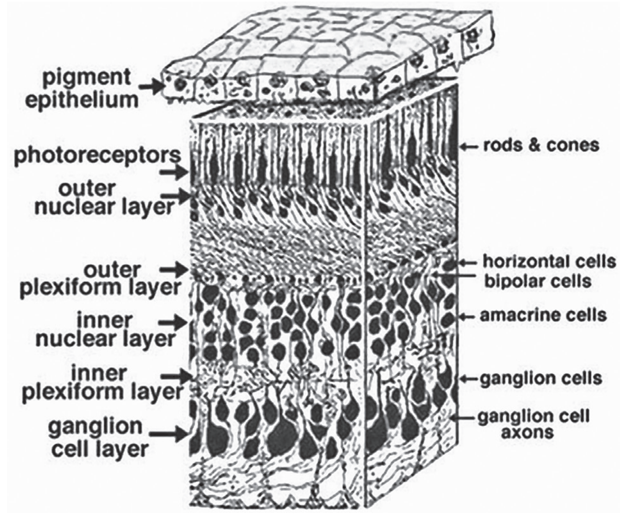

Figure 1. All vertebrate retinas contain layers of nerve cell bodies. Rods and cones are in the outer nuclear layer. (Image republished by permission from Webvision: http://webvision.med.utah.edu/)

Classical photoreceptors — rods and cones — send all photic information to the brain via cells in the outer plexiform layer (OPL) of the retina, which in turn send them to ganglion cells. The visual message leaves the retina as ganglion cell axon spike discharges that occur either when a spot of light stimulates the retina (ON discharge), or when the spot of light is extinguished (OFF discharge). In order to keep the ON and OFF channels separate through the ganglion cells to the brain, the inner plexiform layer (IPL) is divided into two functionally discrete sublaminae, called a (the OFF layer) and b (the ON layer).

Until recently, it was believed that there were only two classes of photoreceptors in the vertebrate retina: rods, which are responsible for vision under very low light levels (scotopic vision), and *cones*, which are responsible for vision at daylight (photopic) levels, color vision, and for discriminating fine details. Rods outnumber cones by a ratio of approximately 20:1.

Photopic vision is generally assumed to occur when the visual system operates at adaptation luminances higher than approximately 3 cd/m^2. Scotopic vision is generally assumed to occur when the visual system operates at adaptation luminances below approximately 0.001 cd/m^2. Rods and cones both operate at light levels between these values (mesopic vision). Although these values for the boundaries between scotopic, mesopic and photopic vision are traditionally cited, the precise values between scotopic and mesopic vision and between mesopic and photopic vision are dependent upon the characteristics of the specific visual scene at any given time.

Rods contain the visual photopigment rhodopsin, with a peak sensitivity at about 507 nm in vivo.[4] The human retina contains short (S-), medium (M-) and long (L-) wavelength cones. The S-cones contain the visual photopigment cyanolabe and are maximally sensitive to wavelengths at about 440 nm in vivo; the M-cones contain the visual photopigment chlorolabe and have a peak spectral sensitivity at about 535 nm in vivo; the L-cones contain the visual photopigment erythrolabe and are maximally sensitive to wavelengths at about 565 nm in vivo.[4] Color vision is initiated in the distal retina by combining the outputs from these three cone types into two spectrally opponent mechanisms, red versus green and blue versus yellow.

Rods and cones are located in the outer nuclear layer of the retina. They respond to a light stimulus with sustained hyperpolarization (increase in the absolute value of the cell's membrane potential), rather than with depolarization (decrease in the absolute value of cell's membrane potential) spikes as exemplified by all other neurons.[2] When a rod or cone photoreceptor is in the dark, potential across photoreceptor membranes is low, the photoreceptor is depolarized, and a steady release of neurotransmitter occurs. When the photoreceptor is illuminated, the potential across the membranes increases and the membrane becomes hyperpolarized, reducing the rate of release of the neurotransmitter.[2]

2.2 Circadian, Neuroendocrine, and Neurobehavioral Photoreceptors

In addition to understanding how the retina converts light signals to neural signals for the visual system, it is necessary to determine the photic pathways involved in circadian, neuroendocrine, and neurobehavioral responses and to characterize which photo-pigments are involved in these responses.

2.2.1 Identifying Photopigments. The methodology to determine the spectral sensitivity of a biological response that depends on an optical radiation stimulus and then establish the underlying photopigments driving the response is called *action spectroscopy*. This technique compares the efficacy of different electromagnetic radiation wavelengths at different irradiances to induce the response.

An action spectrum is one of the principal tools for identifying the photopigments that initiate optical radiation-induced responses. The spectral sensitivity of photopigments is dependent on the amino acid structure of the opsin and the interaction with the chromophore. By definition, an action spectrum is the relative response of an organism to different wavelengths of visible and near-visible electromagnetic radiation.[5-7] Knockout rodent models,[ii] cellular transgenic experiments (transgenic animals are those that have been modified by placing in them, at the embryonic stage, genes of other animals or modified versions of their own genes), and other techniques have been used to try to identify the photopigments concerned with circadian, neuroendocrine, and neurobehavioral phototransduction.

2.2.2 Action Spectra. There are two basic types of action spectra: *polychromatic* and *analytic*. Polychromatic action spectra are developed using broader-bandwith light source stimuli that have half-peak bandwidths greater than 15 to 20 nm, or that emphasize particular wavelengths against backgrounds of "white" light. These types of action spectra are useful for:

- Identifying interactions of biological responses to multiple electromagnetic radiation wavelengths
- Elucidating how organisms respond to optical radiation in more natural settings
- Guiding the development of the more sophisticated analytic action spectra

By themselves, however, polychromatic action spectra have limited utility for identifying the specific chromophores and/or photopigments that initiate photobiological responses[8]

[ii] A knockout is a genetically-manipulated animal (mutant) that may, for example, lack rods and/or cones.

Analytic action spectra, however, in the absence of any confounding factors, should match the absorption spectra of the photopigment driving the response. As in other fields of photobiology, the earliest action spectrum studies on neuroendocrine and circadian responses to light utilized polychromatic stimuli. Polychromatic action spectrum studies in both humans and rodents were reasonably consistent in suggesting that the region between 450 nm and 550 nm provides the strongest stimulation of circadian and neuroendocrine responses (for further review, see Brainard and Hanifin, 2005[9]).

Although diminished in its capacity to elicit neuroendocrine and circadian responses, long wavelength optical radiation at a sufficiently high irradiance can acutely suppress melatonin, as well as phase shift or entrain circadian rhythms in both rodents and humans.[10-14] Similarly, both polychromatic and monochromatic ultraviolet (UV)-A radiation detected by the eyes, although not as strong as wavelengths in the middle of the visible spectrum, can suppress melatonin and regulate the circadian pacemaker, the photoperiodic reproductive response, and the immune response in rodents.[15-17] This physiological regulation is possible because the ocular lens of rodents freely transmits UV radiation (down to 300 nm) back to the retina.

2.2.3 Studies Using Blind Animals. Consistent with the hypothesis that a novel photoreceptor participated in circadian phototransduction, some of the earlier polychromatic and analytic action spectra indicated peak wavelengths that suggested circadian responses might not be mediated by the classic rod and cone photoreceptors.[9] Concurrent to the action spectra work, researchers initiated studies on blind animals and humans to examine whether the lack of classical photoreceptor input to the circadian system interfered with neuroendocrine or circadian regulation. The data emerging from those studies made it increasingly clear that neither the rods nor the cones used for vision were the primary transducers necessary for circadian, neuroendocrine, and neurobehavioral responses.[18-26]

More-recent studies with rodless-coneless mice (*rd/rd cl*)[iii] showed that these mice exhibit normal or even enhanced sensitivity for melatonin suppression and circadian phase shifting of locomotion.[27,28] These results confirm that classical rods and cones are not required for circadian photoreception in mice. Removal of the eyes in both mice and humans, abolishes circadian phase shifting[27,29,30] and melatonin suppression.[19,28] Collectively, the studies suggest that the mammalian eye contains specific photoreceptors for the regulation of circadian, neuroendocrine and neurobehavioral optical radiation responses that are distinct from the classical visual photoreceptors.

2.2.4 The Discovery of Melanopsin. A question remained: what was the photopigment mediating these non-visual responses? In 1998, a seminal discovery was made in the cloning of melanopsin, the fifth opsin-based photopigment, from the mammalian eye. This discovery ultimately led to a paradigm shift in the understanding of mammalian phototransduction. Melanopsin, a molecule found originally in frog-skin melanophore cells (pigment-containing cells on the animal skin that sense and respond to optical radiation), was an early leading contender for the mammalian circadian photopigment, since it is expressed in the ganglion cell layer of the mammalian eye.[31-34] This photopigment shares predicted structural similarities with all known photopigments.

2.2.5 New Photoreceptor Identified. Following the discovery of melanopsin, Berson and colleagues[35] discovered a new class of photoreceptor in the rodent retina; the intrinsically photosensitive retinal ganglion cells (ipRGCs). In contrast to the rods and cones, the ipRGCs are located in the retinal ganglion cell layer, depolarize in response to optical radiation, exhibit a much slower response to an optical radiation stimulus, and have a peak spectral response at approximately 484 nm. Furthermore, the ipRGCs are functionally independent photoreceptors and can still respond to optical radiation even when they are physically or chemically isolated from other neurons.[35-38] **Table 1** outlines the differences between the classical photoreceptors and the ipRGCs.

[iii] Nomenclature used for animals that lack rods and cones (knockout animals).

Table 1: Contrasting Structural and Functional Features of Conventional Photoreceptors (Rods and Cones) and Intrinsically Photosensitive Retinal Ganglion Cells (ipRGCs) of the Mammalian Retina (© Elsevier 2003. Adapted from Berson, 200339 and republished by permission.)

	Rods and cones	ipRGCs
Soma location	Outer nuclear layer	Ganglion cell layer (rarely, inner nuclear layer)
Output	Retina (bipolar and horizontal cells)	Brain (e.g., SCN and OPN)
Light response	Fast hyperpolarizing	Slow depolarizing
Action potentials	No	Yes
Role of retinal pigment epithelium	Essential for photopigment regeneration	Apparently unneccessary for photopigment regeneration
Sensitivity	Moderate (cone) or high (rod)	Low
Receptive field	Very small	Very large
Photopigment	Rhodopsin and cone opsins	Melanopsin?
Photosensitive elements	Outer segment	Soma and dendrites (and axon?)

Abbreviations: ipRGCs, intrinsically photosensitive retinal ganglion cells; OPN, olivary pretectal nucleus; SCN, suprachiasmatic nucleus

The ipRGCs have sparsely branching dendrites (branched fibers that carry signals towards the cell body of a neuron) that are up to several hundred microns long, and most of them terminate in the OFF sublayer of the inner plexiform layer (IPL), described in **Figure 1**. These ipRGCs comprise only 1% to 3% of all rodent retinal ganglion cells; however, because melanopsin is found throughout the dendrites, soma (cell body) and axons, these cells form a diffuse photosensitive net that covers virtually the entire retina.[33] Although ipRGCs respond to optical radiation stimuli very differently from the way the rods and cones do, there is growing evidence that they receive input from rod and cone pathways; more specifically, the ipRGCs receive synaptic input from bipolar and amacrine cells.[40]

This novel photoreceptor is not exclusive to rodents; it also exists in both human and non-human primates. It comprises approximately 0.2 percent to 0.8 percent of all ganglion cells present in the non-human primate retina, and its dispersion of dendritic processes seems to encompass the entire retinal area.[36,41] In the human retina, ipRGCs exist as an extended dendritic tree and form a pan-retinal network.[36,41,42] This broad dendritic arborization is similar to that identified in the rodent retina.[33,37,43]

2.2.6 Short-Wavelength Visible Spectral Sensitivities Characterized. Concurrent with the discovery of melanopsin and ipRGCs, seven recent analytical action spectra have characterized the spectral sensitivity of a range of the physiological responses that are consistent with the short-wavelength sensitivity of these newly characterized sensory cells. Published between 2001 and 2005, action spectra for examined neuroendocrine, circadian, and ocular responses in humans, monkeys, and rodents[35,36,44-49] all showed similar sensitivity to short-wavelength visible (blue) radiation (see **Table 2**).

Predominantly, these action spectra show peak sensitivities in the short-wavelength region of the visible spectrum, with calculated λ_{max} indicating peak photosensitivity of 459 nm to 484 nm. It is important to recognize that the short wavelength sensitivity was first demonstrated prior to the discovery of melanopsin and ipRGCs, when Yoshimura and Ebihara (1996)[26] showed that phase shifting of the locomotor rhythm in *rd/rd* (rodless) mice had a peak response at 480 nm. Given the differences in laboratories, physiological endpoints, animal models, and specific investigative techniques, this consistent detection of peak spectral responses at short wavelengths is remarkably striking (for further review, see Brainard and Hanifin 2005[9]).

In comparing the published action spectra over the past three decades, it is clear that circadian and neuroendocrine researchers have steadily advanced in employing more rigorous photobiological techniques that yield greater consistency:

Table 2: Recent Analytic Action Spectra for Circadian, ipRGC, and Ocular Responses (© Journal of Biological Rhythms, 2005. Adapted from Brainard and Hanifin, 20059 and republished by permission.)

Species	Biological Response	Stimuli Tested	Peak Sensitivity	First author	Year
Human *(Homo sapiens)*	Plasma melatonin suppression	8 fluence-response curves (½ peaks 10–15 nm)	Est λ_{max} = 464 nm (446–477 nm)	Brainard	2001
Human *(Homo sapiens)*	Plasma melatonin suppression	6 fluence-response curves (½ peaks 5–13 nm)	Est λ_{max} = 459 nm (457–462 nm)	Thapan	2001
Mouse *(Mus musculus)*	Pupillary light reflexes	6 fluence-response curves(+/+) (½ peaks #10 nm) (rd/rd cl)	Est λ_{max} = 498 nm or 508 nm Est λ_{max} = 479 nm	Lucas	2001
Human *(Homo sapiens)*	Cone cell ERG-wave	7 fluence-response curves (½ peaks #10 nm)	Est λ_{max} = 483 nm	Hankins	2002
Rat *(Rattus norvegicus)*	ipRGC cellular depolarization	6/10 fluence-response curves (½ peaks 10 nm)	Est λ_{max} = 484 nm	Berson	2002
Mouse *(Mus musculus)*	Circadian phase shift	7 fluence-response curves (rd/rd cl) (½ peaks 10 nm)	Est λ_{max}= 481 nm	Hattar	2003
Monkey *(Macaque nemestrina)*	ipRGC cellular depolarization	10 fluence-response curves (½ peaks 15–20 nm)	Est λ_{max} = 482 nm	Dacey	2005

NOTE: Wild-type and retinally degenerate strains are indicated by (+/+) and (rd/rd cl), respectively. ipRGC=intrinsically photosensitive retinal ganglion cell.

- Fluence (number of photons per unit area) response curves are being developed with monochromatic wavelengths.
- Increased numbers of irradiances are being tested across a larger functional range.
- Greater numbers of wavelengths are being tested across the electromagnetic spectrum.

These improved techniques have increased the accuracy of identifying the estimated λ_{max} for specific responses. A key issue in photobiology is ultimately the matching of physiological responses to absorption spectra of the relevant photopigments. How well do the recently published physiological action spectra compare to the absorption spectrum of melanopsin?

2.2.7 Experiments with Transfected Cells. The only published direct melanopsin absorption spectrum showed that the photopigment is most strongly excited by short wavelength radiation (420 to 440 nm).[50] It is interesting that this peak sensitivity is deeper into the violet-indigo portion of the visible spectrum than the peak sensitivities of the published circadian, neuroendocrine and ocular action spectra. This discrepancy may be due to experimental conditions or the difference of in vitro (when an experiment is performed in a controlled environment outside a living organism) melanopsin responsiveness versus in vivo (when the experiment is performed in or on the living tissue of a whole, living organism) presence in ipRGCs that are closely connected to the filtering of other retinal cells.

Importantly, however, when non-photosensitive cells are transfected with the melanopsin gene (that is, the melanopsin DNA is incorporated in to the non-photosensitive cell), they can become reactive to an electromagnetic radiation stimulus. Specifically, when mouse paraneurons are transfected with the human melanopsin gene they become photosensitive, with a peak sensitivity in the range of 360 to 430 nm.[51-53]

In contrast, human kidney cells transfected with the mouse melanopsin gene become responsive to electromagnetic radiation with peak sensitivity at 479 nm.[54] Similarly, the expression of the mouse melanopsin in *Xenopus* oocytes (immature eggs) confers a peak sensitivity at 480 nm in those cells.[55] Although there is not complete agreement in specific wavelength sensitivity across the studies on melanopsin absorption spectrum and the responses of cells transfected with the melanopsin gene, the studies are consistent in demonstrating a wavelength signature within the short-wavelength spectrum that is distinct from the wavelength sensitivity of rods and cones that mediate vision.

2.2.8 Spectral Sensitivity Inconsistencies. Other inconsistencies in the exact spectral sensitivity of the non-visual responses are apparent in the literature. For example, the action spectrum for pupillary responses of wildtype mice does not identify a λ_{max} in the short-wavelength part of the spectrum but instead identifies a peak at 498 nm to 508 nm.[48] Earlier analytic action spectra with wildtype hamsters and mice also showed peaks in the 500-nm to 511-nm range for circadian phase shifting.[22,26,56] This shift to longer wavelength sensitivity, compared to *rd/rd* and *rd/rd cl* mice, suggests that the intact rodent retina may combine inputs from ipRGCs and classical photoreceptors for at least some of the non-visual responses to optical radiation.[26,47,48,57,58]

2.2.9 Melanopsin Regeneration. Although the states melanopsin passes through during photoexcitation and regeneration are still unknown, studies suggest that melanopsin may form a bistable, photoreversing pigment, most homologous to invertebrate photopigments (e.g., melanopsin most likely evolved from some structure in a common ancestor, such as an invertebrate photopigment).

Opsin-based photopigments are composed of an opsin protein and an optical radiation-detecting chromophore, normally 11-*cis* retinaldehyde. Upon exposure to optical radiation the chromophore is converted into the all-*trans* isoform and the photo-pigment is then in a bleached, inactive form and is dependent upon elements of the retinal pigment epithelium for regeneration to the active form.

In contrast, in the bleached inactive form melanopsin is able to absorb optical radiation and self-recycle back to the active form. This inherent chromophore regeneration capability should increase the efficiency of the photopigment and reduces reliance on external chromophore regeneration tissue such as the retinal pigment epithelium (RPE). When melanopsin is expressed in a system, the presumed photoisomerase activity is enhanced by exposure to optical radiation at 540 nm,[51] implicating long-wavelength sensitivity for this function of melanopsin. Consistently, Tu et al.[180] showed that ipRGC sensitivity in mice with degeneration of rods and cones was not affected by knocking out RPE65, an enzyme that is essential for regeneration of the photoreceptor, nor by acute pharmacological disruption of the retinoid cycle.

One study[59] showed, however, that knocking out RPE65 reduced ipRGC sensitivity as assessed indirectly from pupillary responses. This implies that the ipRGC depends, at least in part, on an extrinsic enzymatic retinoid cascade for its normal function, even though this may be a secondary mechanism. In sum, these studies suggest that more than one mechanism may be involved in melanopsin regeneration.

2.2.10 Circadian Clock Uses Novel Receptor System. Taken together, all of the studies suggest that a novel photoreceptor system is involved in ocular-mediated circadian, neuroendocrine, and neurobehavioral phototransduction. Although full analytic action spectra have yet to be developed, research work has confirmed that shorter wavelength polychromatic[60] and monochromatic[61-63] optical radiation is more potent in humans than exposure to other wavelengths of optical radiation for evoking the same criterion responses for circadian phase shifts, enhancing subjective and objective correlates of alertness, increasing heart rate and temperature, and inducing expression of the circadian clock gene *Per2*.[61,63-65]

Separate lines of evidence support the concept that while the ipRGC's are the central photoreceptors mediating circadian and neuroendocrine regulation, they are not the exclusive photic-detection system. For example, electrophysiological recordings in SCN neurons showed that cones and rods, through bipolar and amacrine cell connections, provide input to the rat circadian pacemaker.[66] Furthermore, three recent studies with melanopsin-knockout mice[iv] demonstrated that although melanopsin plays an integral role in circadian phototransduction and may contribute to pupillary light reflex, it is not necessarily essential for these non-visual ocular mediated responses.[38,43,67] Recent studies in non-human primates have demonstrated that the rods and cones can also activate the ipRGCs through connections to the retina's OPL.[36] Thus, it appears that in primates rod and cone inputs through the bipolar and amacrine cells combine with the inherent photosensitivity of ipRGCs to provide signals for circadian regulation and pupillary regulation.

2.2.11 Key Mediator Role for the ipRGCs. Psychophysical data show a peak spectral sensitivity for melatonin suppression in humans shorter than the peak sensitivity of the ipRGCs.[44,49] A series of studies testing melatonin suppression in humans using polychromatic and monochromatic light sources suggested that the visual photoreceptors work in combination with the ipRGCs to mediate this response in humans.[62,68-72] Recent work[73] has suggested that the response to polychromatic white light sources cannot be predicted from the melanopsin photosensory spectral sensitivity, and that either the melanopsin photoreversal function or the visual photopigments have an influence on the melatonin suppression response.

It is important to note, however, that depending upon the combination of wavelengths in polychromatic light sources and their absolute irradiances, the relative contribution of photopigments, and thus the spectral response, can be different.[70-72] Two of these studies[68,69] suggest that, similar to color vision, human circadian phototransduction exhibits spectral opponency; that is, some polychromatic white light sources are weaker stimuli to the circadian system than would be assumed from the sum of the effects produced by their individual spectral components.

These two studies also suggest an S-ON participation in human circadian phototransduction. An S-ON response means that bipolar cells depolarize in response to light and thus increase neurotransmitter release from the cell's terminals (ON response); an S-OFF response means that they hyperpolarize in response to light and thus decrease neurotransmitter release (OFF response) from the cell's terminals. Recently, Wong et al., 2007[74] showed that ON bipolar and amacrine cells make the most direct contributions to the ipRGC responses, while OFF bipolar cells make a very weak contribution. Primate ipRGCs also project to the LGN, the principal relay center to the visual cortex, suggesting that ipRGCs may play a role in higher visual processing in diurnal primates.[36] The retrograde cells they measured from the LGN had an S-OFF response.

While it is becoming increasingly clear that the visual system contributes significantly to circadian, neuroendocrine, and neurobehavioral responses to optical radiation, a short-wavelength non-rod, non-cone photoreceptor — presumably the ipRGC — is the primary mediator of these responses.

3.0 Overview of the Circadian, Neuroendocrine, and Neurobehavioral Responses to Optical Radiation

Optical radiation does not just impact the visual system; it also has effects on a range of neuroendocrine and neurobehavioral responses, including the circadian system. The world is a highly rhythmic place, with predictable environmental changes occurring on both an annual (seasons) and a daily (day/night) basis. As a result, organisms have evolved endogenous circadian clocks that drive rhythms in physiology and behavior, anticipating daily environmental time cues. These internal clocks ensure that physiological and behavioral events are appropriately coordinated, both externally (with the environmental cycle), and internally (with each other).

[iv] Genetically-manipulated mice that do not have melanopsin.

This synchronization increases an organism's chances of survival; it also ensures efficient metabolism since behavioral and physiological systems will be functioning maximally only when required. Biological clocks and the rhythms that they drive are classified according to their period (the time taken for one cycle to be completed). The most prevalent rhythms are daily oscillations having approximately 24-hour periods.

3.1 Resetting the Circadian Clock

Circadian clocks do not run at exactly 24 hours.[75] Environmental time cues must be able to reset this internal clock to ensure that physiology and behavior are appropriately synchronized with the outside world. The major environmental time cue able to reset (phase-shift) these rhythms is the 24-hour light-dark cycle (for a review see Czeisler and Wright 1999[76]), although non-photic time cues such as exercise, food intake, social interactions, and sleep may also affect the circadian pacemaker in humans.[77-81]

Optical radiation information is captured by retinal photoreceptors, converted into neural signals and conveyed directly to the SCN of the anterior hypothalamus (the site of the endogenous pacemaker) via a dedicated neural pathway, the RHT.[3,82,83] The 24-hour light-dark cycle resets the internal clock on a daily basis; in turn, this clock signals a wide range of brain areas, resetting clock-controlled physiology and behavior.

For synchronization with the environment (entrainment) to occur, the circadian clock's sensitivity to the resetting stimulus must change periodically. This allows phase shifts having different direction and size to be induced depending on the characteristics of the administered stimulus.[84,85] Multiple optical radiation characteristics (quantity, spectrum, timing, duration, pattern, and prior optical radiation exposure) all affect the magnitude of the phase-resetting response. The following sections will outline these properties in more detail.

3.2 Desynchronized Pacemaker in Blind Individuals

The vital importance of optical radiation input to the circadian pacemaker is readily observed when the circadian rhythms of totally blind individuals, particularly those who are bilaterally enucleated, are examined.[30,86-89] In such people, the sleep-wake cycle, alertness and performance patterns, and the rhythms of temperature and some hormones become desynchronized from the 24-hour day as a result of optical radiation information failing to reach the SCN to synchronize the clock and its outputs.[30,90,91] In these cases, the circadian pacemaker runs at its own endogenous period, ranging from approximately 23.6 to 25.1 hours in blind people,[30,86-89] and is not reset to 24 hours on a daily basis.

The resulting clinical condition—non-24-hour sleep-wake disorder—is observed in the majority of totally visually blind people and is characterized by a cyclic sleep disorder with episodes of good sleep followed by episodes of bad sleep and excessive daytime naps as the internal pacemaker runs in and out of synchrony with the 24-hour day.[90,92] The impact of loss of regular and appropriately-timed optical radiation input in sighted subjects was initially explored in long-term cave experiments and then more recently in sophisticated laboratory studies.[75,93-97] These studies showed that without appropriately timed daily optical radiation information, the circadian clock does not remain entrained to the 24-hour environmental cycle.

3.3 Photic Signal Pathways

In mammals, the eye is responsible for detecting optical radiation for the clock[21,30] and this photic signal is primarily transmitted from the retina to the clock (situated in the SCN of the hypothalamus) via a direct pathway called the RHT.[98] The RHT is formed from the axons of a distinct, relatively small set of retinal ganglion cells (RGCs) that are broadly distributed throughout the retina.[83,99] This wide distribution presumably allows them to provide an integrated photic input from a large area of the retina and therefore information about overall environmental irradiance.

The integrity of the RHT is essential for the entrainment of circadian rhythms by the light-dark (LD) cycle.[100] The RHT also projects to many extra-SCN sites in the brain,[101] potentially mediating a range of optical radiation responses including circadian, neuroendocrine, and neurobehavioral responses.

=A second photic pathway to the SCN comes from the geniculo-hypothalamic tract (GHT), which arises from a lamina of retino-recipient cells in the IGL situated between the ventral lateral and dorsal LGN of the thalamus.[102-106] The GHT projects bilaterally and primarily to the same subdivision of the SCN as the RHT, namely the ventrolateral region, and has terminals that overlap with those of the RHT.[102,107] GHT integrity is not essential for the photic entrainment of rhythms because IGL lesions do not stop entrainment to LD cycles.[108,109] Such integrity may play a large modulatory role in photic entrainment, however, as IGL lesions do cause changes in circadian rhythmicity. Lesions slow the rate of re-entrainment after phase-shifts, reduce the free-running period, and reduce the magnitude of advance phase-shifts.[109-111]

3.4 Organs with Peripheral Clocks

Until a few years ago it was thought that the SCN was the exclusive mammalian clock. Circadian organization perceptions were altered, however, with the discovery of peripheral clocks in various organs including the liver and lung.[112] It is thought that these peripheral clocks are responsible for local time-keeping within a tissue/organ and drive rhythmicity in tissue-specific genes and processes. These clocks, however, can exhibit self-sustaining circadian oscillations, but in the absence of the SCN they are not able to retain synchrony externally with the 24-hour day or internally with each other. The SCN is the only “clock” capable of generating self-sustaining, endogenous oscillations, and thus it is the master circadian oscillator.

It appears that the SCN drives and sustains rhythmicity in peripheral clocks while also entraining them to an appropriate phase relative to each other and the external environment (for a recent review see Panda and Hogenesch, 2004[113]). In addition to acting as a peripheral-clock coordinator, the SCN directly drives behavioral rhythms via diffusible, humoral signals, as well as oscillations in other brain areas. In turn these oscillations can influence endocrine or autonomic systems via direct or multisynaptic (pertaining to or relaying through more than one synapse) neural connections.[114]

3.5 Circadian Clock Markers

In mammals, a wide variety of physiological and behavioral events exhibit circadian rhythmicity, ranging from the obvious sleep-wake cycle to more covert changes in hormone levels, core body temperature, blood pressure, and gene expression. Perhaps the most pertinent circadian rhythms for the purpose of this Technical Memorandum are those which can be used as markers of the phase (timing) of the clock and hence reveal the impact of optical radiation stimulus on the clock.

Core body temperature (CBT) and the pineal hormone melatonin are the most commonly used clock markers. Melatonin is the most popular phase marker, as it is not subject to as many masking influences as CBT and it can be measured non-invasively. The SCN drives the circadian rhythm in pineal melatonin production (i.e., high melatonin levels at night and low melatonin levels during the day) via a multisynaptic pathway that projects to the paraventricular nucleus of the hypothalamus (PVN) and the superior cervical ganglion (SCG) (reviewed in Klein et al., 1991[115]).

4.0 Lighting Characteristics Affecting the Visual System

Luminous efficiency functions are used by the lighting industry to relate human spectral sensitivity to radiant power at different wavelengths. The *photopic* luminous efficiency function [$V(\lambda)$], as defined by the International Commission on Illumination (CIE), is nearly universally used in photometry. This function is employed in the fundamental definition of luminous flux and other photometric quantities in the conversion of radiometric to photometric quantities.

The CIE photopic luminous efficiency function is based on the relative amounts of power needed at each wavelength to produce a criterion brightness response in a 2° foveal field of view, and peaks at 555 nm. The current definition of light is based solely on the visual response of humans; other species are sensitive to different parts of the electromagnetic spectrum. These other species have different photopigments for converting radiant power into neural signals for vision.

Geometry and spectrum define photometry. The geometric characteristics of light establish the spatial units typically used in photometry:

- Luminous flux incident on a surface area is called *illuminance* (flux per element area).
- Luminous flux within a solid angle is defined as *luminous intensity* (flux per solid angle).
- Luminous flux distributed within a solid angle from a planar element is called *luminance* (luminous intensity per element area).

The spectral characteristics of light weight radiant flux according to the *photopic* luminous efficiency function [$V(\lambda)$] (representing a weighted linear combination of L and M human cone photoreceptors). In rare cases the *scotopic* luminous efficiency function [$V'(\lambda)$] (representing the human rod photoreceptors) is used to weight radiant energy, but it should be recognized that if electric light is present it is unlikely that the scotopic response will be dominating visual function.

The range of light characteristics that affect visual function include quantity, spectrum, timing, duration, and spatial distribution. Briefly outlined next are how these characteristics affect vision. For more in-depth discussion on the lighting characteristics affecting the visual system, the reader is referred to *The Lighting Handbook*. (*Note:* these characteristics may be quite different from those identified as maximally effective for circadian, neuroendocrine, and neurobehavioral phototransduction in **Section 5.0**).

4.1 Quantity

The human visual system can process information over a large range of luminances (about 12 log units), though not all at once. In order to cope with a range of retinal illuminances from those of outdoor very dark nights (scotopic vision) to outdoor daytime light levels (photopic vision), the human visual system changes its sensitivity continuously through the different absolute sensitivities of rods and cones, a process called adaptation.

4.2 Spectrum

The process of adaptation also influences the spectral sensitivity of the visual system because different combinations of photoreceptors are operating at different retinal illuminances. At photopic light levels (traditionally, adaptation luminances higher than about 3 cd/m^2), retinal visual response is dominated by cone photoreceptors in the fovea and periphery. The relative luminous efficiency function for the CIE Standard Photopic Observer has a maximum sensitivity at 555 nm.[116,117]

At scotopic light levels (traditionally, adaptation luminances below about 0.001 cd/m^2), only rod photoreceptors respond to light stimuli, and the fovea (which contains only cones) is not operating. The relative luminous efficiency function for the CIE Standard Scotopic Observer has a maximum sensitivity at 507 nm.

At mesopic light levels, (traditionally, adaptation luminances between 0.001 and 3 cd/m^2), both rod and cone photoreceptors are active for vision. There is no official relative spectral sensitivity curve for the mesopic state, although Rea et al. have recently proposed a unified system of photometry that bridges the photopic and scotopic luminous efficiency functions through the mesopic region.[118]

4.3 Timing

Although the underlying physiology is still unclear, it is known that visual sensitivity, and thus visual discrimination threshold, change during the 24-hour day/night cycle. This sensitivity is low in the morning and increases progressively over the course of the day, reaching a first peak at 22:00 hours. A similar pattern occurs during the night, with low threshold, levels at the beginning and high levels when night ends. Despite these changes in absolute sensitivity, the visual system is always primed and will exhibit sensitivity to light at any time of the day or night.

4.4 Duration

The visual system not only responds to variations of luminance in space, but also responds to luminance variations in time. The critical flicker (or fusion) frequency (CFF) is that point where a repetitive flashing stimulus is perceived as steady, rather than intermittent.

A better way to characterize the combined effects of luminance modulation and temporal frequency is by using the temporal modulation transfer function (MTF). Like the CFF, the MTF defines the thresholds for vision at different temporal frequencies. (For more in-depth discussion about CFF and MTF, please refer to *The Lighting Handbook*). In general, the visual system operates very quickly (within a few hundred milliseconds) to suprathreshold flashed or moving targets.

4.5 Spatial Distribution

Spatial distribution of light is critical to vision. Through optical refraction by the cornea and lens in the eye and by neural-optical enhancements in the retina, the spatial distribution of objects and textures in the environment can be very accurately processed by the visual system. Precise registration of spatial information on the retina is imperative for processing visual information.

Target visibility can be defined in terms of contrast sensitivity functions (CSFs) and suprathreshold visual performance. CSFs define the threshold for vision at different spatial frequencies (sizes). *Acuity* is the highest spatial frequency that can be seen with a contrast of one.

Among the other factors that affect contrast sensitivity function, adaptation luminance is of particular importance. As adaptation luminance increases, the contrast sensitivity increases for all spatial frequencies, the spatial frequency at which the peak contrast sensitivity occurs increases, and the highest spatial frequency that can be detected increases (reviewed in Rea, 2000[119]). Suprathreshold visual performance describes how well an observer can see a target rather than whether the target can be seen, as is the case for the CSFs. Similar to contrast sensitivity, adaptation luminance affects suprathreshold visual performance. As the adaptation level increases, the speed and accuracy of visual processing continues to improve.

5.0 Optical Radiation Characteristics Affecting the Circadian, Neuroendocrine, and Neurobehavioral Systems

As reviewed in **Section 4.0**, the eye performs a dual role in detecting optical radiation for a range of behavioral and physiological responses separate and apart from sight. While optical radiation reaching the retina will provide visual sensation, it is also the primary stimulus that regulates the timing of various biological parameters.

Currently, there is no formal definition of the geometric and spectral characteristics affecting the non-visual phototransduction, although a large body of work has explored how various properties of optical radiation affect these responses. These are outlined in this section (for further reviews, see Czeisler and Wright, 1999, and Brainard et al., 1997[76,120]). It is important to note that large (orders of magnitude) differences exist in many properties of non-visual photoreception as compared to vision. This invites much questioning about the currently considered fundamental characteristics of optical radiation perception in humans.

5.1 Quantity

More than two decades ago, it was demonstrated that optical radiation could affect the circadian system but it was thought that bright light (approximately 2,500 lux [approx. 250 fc] vertical at the eye, which is about 7,500 to 10,000 lux on the horizontal surface) was required and that dim light (100 lux [10 fc] at the cornea, which is about 300 to 500 lux on the horizontal) was ineffective.[121-123] Soon after those initial discoveries, however, it was demonstrated (under tightly controlled laboratory conditions) that exposure to between 1 lux and 5 lux (0.1 fc and 0.5 fc) at the cornea of specific monochromatic wavelengths of optical radiation (460 nm and 509 nm, respectively) could suppress melatonin in healthy humans,[44,124] suggesting that the system was much more sensitive than initially thought. As will be discussed in this section, it is not appropriate to use illuminance to characterize optical radiation for the circadian system.

Subsequent laboratory work to determine the sensitivity threshold of the circadian system has demonstrated that the human circadian pacemaker phase-shifts in response to relatively low levels of a broadband spectrum white light source (approximately 100 lux [10 fc] at the cornea or 300 to 500 lux [30 to 50 fc] on the horizontal plane).[125,126] In fact, dose-response curves for a single 6.5-hour exposure of 9,500 lux (950 fc) of a white light source (4100-K fluorescent lamp) during the biological night, centered 3.5 hours before minimum core body temperature, show a sigmoidal function. This indicates that the phase-delay resetting response saturates at approximately 600 to 1,000 lux (approx. 60 to 100 fc) at the cornea, with approx. 100 lux at the cornea generating about 50 percent of the maximum resetting response (see **Figure 2**).

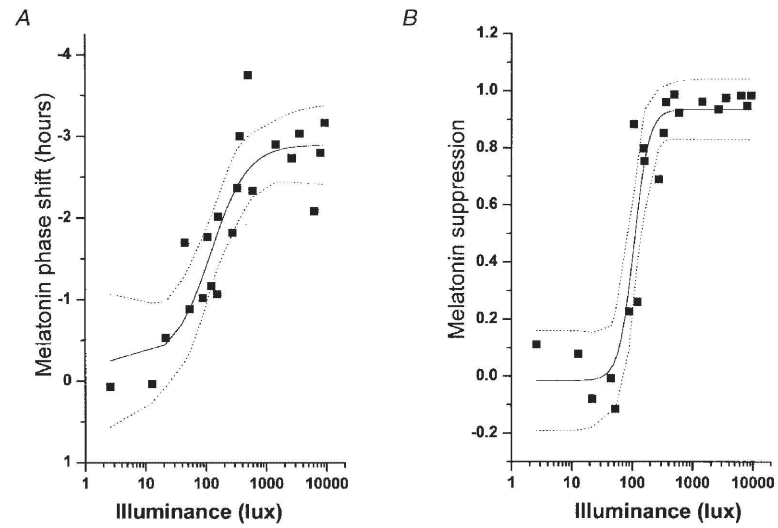

Figure 2. Dose-response curves for phase resetting (A) and melatonin suppression (B) in response to a single 6.5-hour exposure of white light at the cornea (4100-K fluorescent lamp) during the biological night centered 3.5 hours before minimum core body temperature. (©Blackwell Publishing 2000. Reproduced from Zeitzer, 2000126 and republished by permission.)

Earlier studies demonstrated that exposure to a white incandescent light source could suppress melatonin at 100 lux at the cornea of subjects with dilated pupils.[120,127] Jasser and colleagues,[71] however, showed no melatonin suppression after 90-minute exposure to 18 lux (1.8 fc) at the cornea of subjects with dilated pupils.

A similar dose-response (see **Figure 3)** was also observed for the acute effects of optical radiation on subjective alertness, attention failures, and EEG-derived changes in brain activation.[128]

Other studies conducted in laboratory settings failed to show circadian entrainment to 12h:12h, 200:<8 lux light-dark cycles without scheduled sleep or activity (in this experiment, subjects were exposed to 12 hours of a light source providing 200 lux [20 fc] at the cornea and 12 hours of less than 8 lux [0.8 fc] at the cornea).[129] Wright et al.[130] showed, however, that if a very rigid light-dark/activity-rest pattern is imposed in a subject, circadian entrainment to a very dim light (approximately 1.5 lux [0.15 fc] at the cornea) can be achieved. The authors of the study were

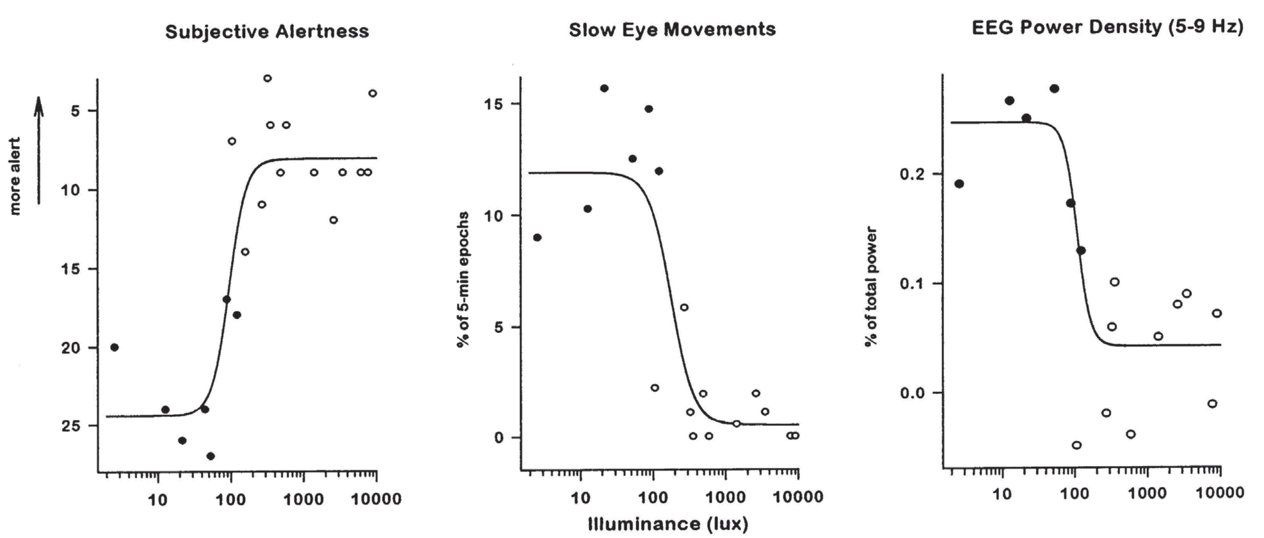

Figure 3. Dose-response curves for subjective alertness, attention failures, and EEG-derived changes in brain activation after exposure to optical radiation at the cornea (4100-K fluorescent lamp) during the nighttime. (© Elsevier 2000. Adapted from Cajochen, 2000128 and republished by permission.)

unsure whether the entrainment was a result of the optical radiation exposure or the potential role of non-photic stimuli, such as the sleep/wake cycle, in this schedule.

While non-photic time cues can entrain the pacemaker under some conditions,[80] and very dim light cannot maintain entrainment to a schedule requiring a larger daily phase shift (e.g., 23.5 or 24.6 hours[130]), scheduled sleep-wake cycles in the absence of optical radiation are insufficient to maintain entrainment.[30,86,88] Based on these studies, it appears that levels of optical radiation lower than originally demonstrated by Lewy and colleagues,[123] and as low as 50 to 100 lux (approx. 5 to 10 fc) at the cornea, can impact the circadian system of humans in laboratory settings.

Levels of optical radiation required to impact the circadian clock outside of laboratory conditions are still unknown, largely because such studies have not been extensively conducted. One study,[131] which measured optical radiation using a photosensor around the subjects' neck as a medallion, suggested that levels as low as 40 lux (4 fc) measured on a vertical surface (which was about 200 lux [20 fc] on the horizontal surface) for three hours in the evening (between 22:00 and 01:00 hours) may have induced a 0.6-hour phase delay of the subjects' circadian pacemakers.

The minimum levels of optical radiation necessary to impact the circadian clock in the real world will likely differ from those observed under laboratory conditions. Photic history has a profound effect on system sensitivity[70-72] in that there is increased sensitivity following an interval of dim lighting. Thus, it might be predicted that higher levels of optical radiation would be required to induce a response following a period of high light levels (i.e., daylight). Consistently, visually-impaired subjects with the lowest acuity level (having only the ability to differentiate between light and dark) remain entrained to the 24-hour LD cycle.[30]

5.2 Spectrum

As just discussed, information regarding the spectral sensitivity of the human circadian system has emerged only recently. But it is now widely accepted that sensitivity peaks in the short-wavelength portion of the visible spectrum, and that multiple photopigments have the capacity to participate in circadian phototransduction. Several groups have demonstrated the sensitivity of circadian and other acute responses to short-wavelength optical radiation.[44,46,49,60-65,68,69,132-135] But insights into the circadian system's spectral sensitivity were also based on a series of studies conducted using animal models (see **Section 3.0**).

Although the use of knockout models is essential to identify which photopigments are necessary for a response, it is also vital to determine the spectral sensitivity of the circadian system in normal animals. This sensitivity may differ markedly from that observed in knockout animals because it will reflect the input of all the photopigments involved.

For example, in rodless-coneless laboratory mice, the action spectra for both phase shifting and PLR have a λ_{max} of approximately 480 nm,[47,48] reflecting phototransduction by melanopsin, whereas in wild type mice[v] these action spectra are shifted to longer wavelengths (λ_{max} to approximately 500 nm) and exhibit a wider peak,[26,48,57,58] presumably indicating input from more than one photoreceptor (melanopsin ipRGCs, rods, cones).

It is entirely possible that circadian responses in diurnal humans having a rod and tri-chromatic cone visual system may exhibit a different spectral sensitivity than nocturnal rodents with their distinct visual system (rods, UV-cones, and M-cones). In humans it is obviously much more difficult to determine the specific photopigments involved and their relative contributions. It is clear, however, that the spectral sensitivity of non-visual phototransduction differs significantly from that required for vision (e.g., acuity, brightness, visual performance, and color).

Recent studies suggest that melanopsin and classical photopigments participate in the circadian phototransduction of various species, from the mouse to the primate/human. Consistently, studies of blind individuals with no light perception (no measurable visual response) have shown that a small proportion of such subjects are capable of melatonin suppression and circadian phase-shifting.[19,29,136] Furthermore, red-green color-blind subjects exhibit a normal melatonin suppression in response to white or green light,[137]

[v] Wild type mice are not knockout mice.

suggesting that functional rods and/or cones are not necessary for non-visual photic responses.

The wavelength where normal humans exhibit maximum non-visual sensitivity should also be considered when designing architectural lighting. Similarly, the wavelength sensitivity of different species will determine the optimum environmental lighting for these animals.

5.3 Timing

Crucial in determining the direction and magnitude of circadian phase-resetting effects is the timing of any optical radiation exposure. Exposure at one time of day can shift the circadian pacemaker timing earlier (i.e., advance the clock phase); exposure at another time of day can shift the pacemaker timing later (i.e., delay the clock phase).

The change in direction and magnitude of the phase shift as a function of time of exposure to optical radiation can be plotted as a Phase Response Curve (PRC).[96,138-144] A diagram representing the human PRC to optical radiation for someone living under normal LD conditions is shown in **Figure 4**. The phase shifting effects of optical radiation (ordinate) to either a later time (phase delay, negative value) or an earlier time (phase advance, positive value) are plotted against the time of day of exposure (abscissa).

An eight-hour sleep episode is superimposed from 0:00 to 8:00 hours. Under normal conditions, optical radiation exposure between 18:00 to 6:00 hours (before the minimum core body temperature is reached) causes a pacemaker phase delay, with a maximum delay at about 2:00 a.m. Optical radiation delivered between 6:00 to 18:00 hours (after the minimum core body temperature is reached) causes the clock to advance, with a maximum advance occurring after exposure in the morning (approx. 9:00 hours). Note that the timing of minimum core body temperature can vary among individuals.

Optical radiation exposure has a maximum effect in shifting the pacemaker when it occurs during the biological night. This is when humans are usually asleep and therefore normally encounter minimum light. Exposure is less effective during the biological day.[143,144] In most animal PRCs there is a “dead zone,” a time during the biological day when optical radiation exposure is ineffective for inducing phase shifts. However, in many human PRCs, including the most comprehensive human PRC published to date,[141] there is no evidence of a dead zone.

The example in **Figure 4** is a simple human PRC and would be the expected response to a single exposure of long duration (e.g,. up to seven hours) with a moderate light level (up to 10,000 lux [approx. 1,000 fc]). Specifically, this is termed a weak Type 1 PRC and predicts that the maximum phase shift obtained will be less than 12 hours. A strong Type 0 PRC can also be produced, which predicts a maximum phase shift of 12 hours or more. In humans, three consecutive daily five-hour exposures to 10,000 lux at the cornea can generate a Type 0 PRC producing a net phase shift of 12 hours.[138, 140]

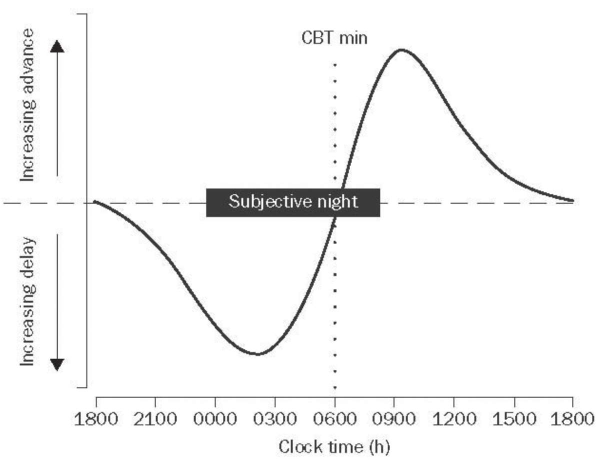

Figure 4. Type 1 phase response curve to optical radiation. (© Elsevier 2001. Adapted from Rajaratnam and Arend, 2001[145] and republished by permission.)

It has been suggested that such large phase shifts occur via suppression of the circadian pacemaker amplitude to near zero, essentially "stopping the clock" before immediately shifting it to a new phase up to 12 hours different.[140] While the currently-available PRCs provide information on general optical radiation timing effects, further studies are required on the interaction of this timing with the quantity, duration, distribution, and wavelength of optical radiation.

5.4 Duration

The phase-shifting effects of optical radiation are also dependent on the duration and pattern of optical radiation exposure and vary exponentially with duration.

A daily three-hour exposure to 5,000 lux (approx. 500 fc) at the cornea was as effective as a six-hour exposure for adaptation to an experimental night shift.[146] The PRC for one-hour exposure to 10,000 lux at the cornea from a polychromatic light source has approximately 45 percent of the PRC amplitude for a 6.7-hour exposure to the same amount of optical radiation.[141,147] Intermittent, as opposed to continuous, exposure induces a greater phase shift than predicted by a simple linear response to optical radiation duration.[148,149] Rimmer and colleagues showed that intermittent exposure to high levels of optical radiation at 25- or 90-minute intervals (representing exposure during 63 percent or 31 percent, respectively, of a five-hour period), caused 90 percent and 70 percent of the response when compared to a continuous five-hour exposure to 9,500 lux (approx. 950 fc) at the cornea.[149] Similarly, compared to a single 6.5-hour continuous exposure, intermittent exposure (23 percent of the total duration) composed of six 15-minute cycles of 9,500 lux white light exposure with 60 minutes of dim light (less than 1 lux at the cornea) in between each cycle, caused 77 percent of the continuous exposure response.[148]

These data suggest that the degree of phase shift increases exponentially with optical radiation exposure duration, and that the first half of the exposure induces a disproportionately higher effect than the latter half. This function, when combined with an understanding of how duration interacts with optical radiation level and wavelength, may allow the development of lighting regimes that induce the maximum circadian phase reset with the least possible exposure duration.[150-152]

5.5 Spatial Distribution

Unlike the visual system, non-visual photoreception does not require precise spatial resolution of optical radiation because it is concerned with changes in ambient irradiance. The distribution and number of ipRGCs mediating these non-visual responses support this hypothesis. Non-visual receptors consist of a small number of the total retinal ganglion cells, spread nearly uniform across the retina in a net-like distribution. These cells also have very large dendritic fields that are photosensitive, which further assists broad (but relatively insensitive) optical radiation detection.[33,35]

It has been shown, however, that optical radiation coming from the upper visual field (and reaching the lower part of the retina) is more effective in suppressing melatonin than the same amount of radiant flux coming from below the horizon and stimulating the upper part of the retina.[153,154] Other studies suggest that exposing the nasal portion of the retina evokes a stronger melatonin suppression than exposing the temporal portion.[155,156]

5.6 Adaptation (Photic History)

Evidence is emerging that the human circadian system's sensitivity to optical radiation may be determined by optical radiation exposure over the hours (and possibly days) immediately preceeding,[70-72] with non-visual phototransduction exhibiting an adaptation response. The importance of relative changes in optical radiation level, rather than the absolute level, has been known for several decades from animal studies[157,158] in relation to the photoperiod effects of optical radiation.

Photic history (from the preceding days and weeks) also influences human sensitivity to optical radiation at night as measured by melatonin suppression. The higher the exposure to optical radiation during the day (e.g., one week of exposure for four hours per day to outdoor lights), the lower the human circadian system's sensitivity becomes to optical radiation at night (measured by nocturnal melatonin suppression in Hebert et al.[70]). Even as little as two hours' exposure to dim white incandescent light at 4 cd/m^2 can significantly dampen the subsequent melatonin suppression response to blue light in healthy subjects.[71] This property has also been recently demonstrated for the ipRGCs themselves, where the photic response magnitude was shown to adapt as a function of prior exposure irradiances.[159]

Wong and colleagues[159] showed that the background-evoked response of the ipRGCs decayed and the responses to a superimposed light flash increased after exposure to a consistently bright background. Once this background was removed, ipRGC sensitivity recovered, suggesting that these cells go through both light and dark adaptations. The mechanism by which this adaptation occurs is not yet known (for example, unlike rod and cone opsins, melanopsin does not appear to bleach). However, these findings demonstrate that the circadian system more likely responds to *changes* in optical radiation levels, rather than absolute exposure.[160]

6.0 Measuring Optical Radiation for Circadian, Neuroendrocrine, and Neurobehavioral Regulation

Generally, any characterization of optical radiation exposures for photobiological responses should include the radiometric quantification of quantity and spectrum.[161,162] Direct measurement is commonly given as photon density or irradiance determined as a function of wavelength. When describing a light source with a narrow spectral bandwidth, the total power per unit area (or total flux per unit area) is sufficient to quantify the optical radiation stimulus. But when the spectral bandwidth of the optical radiation stimulus increases, this is not an adequate measure because photobiological responses vary significantly in their sensitivity to different wavelengths.

A numerical characterization of wider-bandwidth optical radiation can be calculated by weighting the spectral values with a spectral weighting function appropriate to the effect under consideration. For example, there are standard, defined spectral weighting functions for rods and cones that apply when characterizing light for visual responses,[119,163] but there are no standard spectral weighting functions for circadian and neuroendocrine responses.

The human eye detects non-visual information about optical radiation in addition to supporting vision and visual reflexes. When the acute suppression of melatonin was first observed,[123] there were no defined action spectra for circadian or neuroendocrine regulation in humans. As a consequence, photopic measures of light (in lux) were often used as a surrogate measure during human studies of circadian and neuroendocrine physiology. In practical applications, recommended doses of optical radiation for therapeutic purposes have often been provided in terms of a given illuminance (in lux) at a specified distance from the eye.[164-168]

Human studies past and present have specifically demonstrated that using illuminance for characterizing the optical radiation involved with melatonin suppression and circadian phase shifting is inappropriate.[17,62,124] Specifically, those action spectra show that (relative to the photopic luminous efficiency function, based upon L-cone and M-cone responses) the peak spectral sensitivity for melatonin regulation is in the short-wavelength portion of the visible spectrum.

More-recent studies, based on selected wavelength comparisons, have indicated that circadian phase-shifting and the acute alerting effects of optical radiation also peak in the short-wavelength spectrum for healthy humans.[60-63,65] Similarly, an initial clinical trial of light therapy for winter depression (a.k.a. seasonal affective disorder, or SAD) indicates that narrow-bandwidth blue light may be particularly potent for treating this condition.[169] Much work remains to be done, however, before there is any certainty about the optimum blend of wavelengths best suited to such applications.

Both older and more recent studies on enhanced sensitivity in the short-wavelength portion of the visible spectrum for acute melatonin suppression, acute alerting effects, circadian phase-shifting, and treating winter depression strongly indicate that the current practice of using photometric values (in lux or fc) to quantify such therapeutic light for SAD, sleep disorders, or circadian disruption (from shiftwork or jet travel) is not optimal. The need for a specific optical radiation measurement technique applied to human circadian regulation and light therapy has been known for years[44,49,124,170,171] and is being actively discussed within the lighting engineering community.[170-178]

Ultimately, the way optical radiation is described and measured for circadian, neuroendocrine, and neurobehavioral responses will need to be redefined around a new, dedicated metric that can best quantify optical radiation for such responses. Preliminary models have been developed,[173-175,179] but the proposed systems do not yet incorporate all key aspects of the optical radiation stimuli (timing, duration, quantity, wavelength, pattern, and history). Developing such a metric should not be the work of a single laboratory, but rather should involve a broad collaboration across the circadian, neurobehavioral, visual science, and illuminating engineering communities.

7.0 Conclusions

This Technical Memorandum was developed to discuss the basic aspects of and latest information about biological systems affected by optical radiation reaching the human retina. Optical radiation detected by the retina impacts the individual's behavior, psychology, and perception of the world environment. Thus far, most published scientific research in this area has been based primarily on controlled laboratory and clinical experiments. Some studies have attempted to implement the basic research results into practical situations where they could prove beneficial. Therefore, it is premature to make design recommendations in this Technical Memorandum. Once more field studies investigating the impact of optical radiation on human health and well-being are conducted and the robustness of optical radiation's effects outside of laboratory conditions have been demonstrated, the Light and Human Health Committee will report on the practical implications. This will include a discussion of new applications being adopted by lighting practitioners, descriptions of emerging technologies, and identification of unresolved problems that require further research and development. The goal remains a more complete understanding of human visual, circadian, neuroendocrine, and neurobehavioral responses leading to improved designs for all lighted environments.

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